RAG Learning

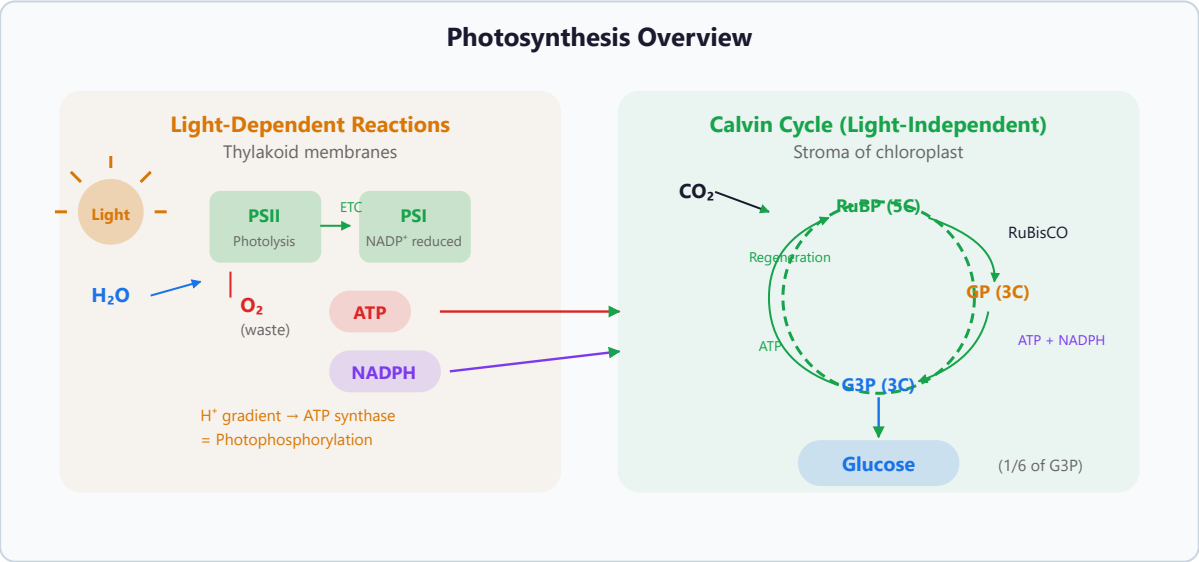
S5 — On the Wild Side

5.1 Photosynthesis

Overview of Photosynthesis

Photosynthesis: The process by which plants, algae, and some bacteria convert light energy into chemical energy (stored in glucose), using carbon dioxide and water as raw materials and releasing oxygen as a by-product.

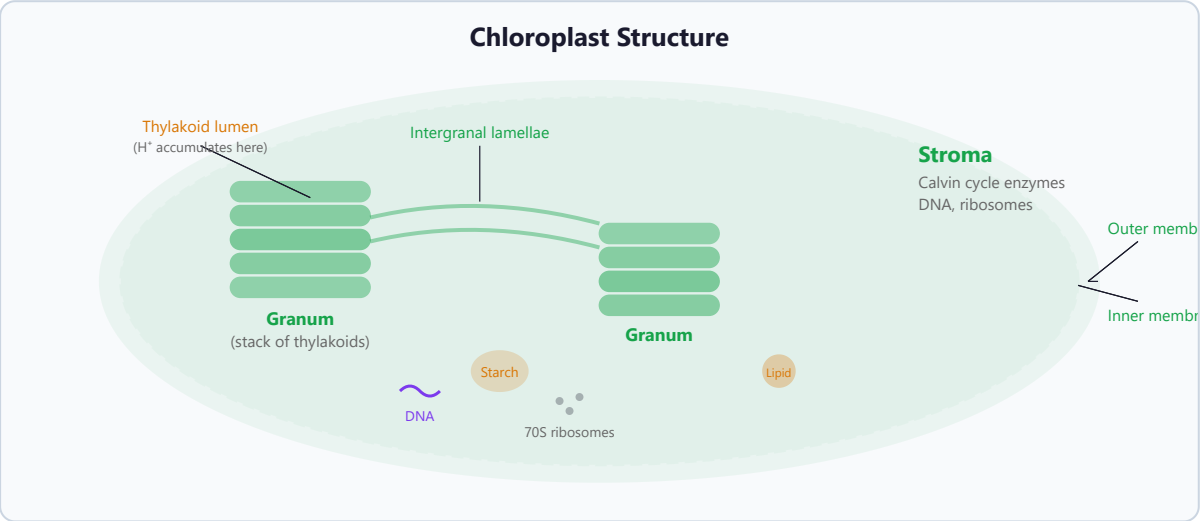
$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$



Photosynthesis occurs in two stages: the light-dependent reactions (in the thylakoid membranes) and the light-independent reactions (in the stroma of chloroplasts).

Chloroplast Structure

Chloroplasts are surrounded by a double membrane (envelope). Internally, they contain an extensive system of thylakoid membranes arranged in stacks called grana (singular: granum), connected by intergranal lamellae. The thylakoid membranes contain photosynthetic pigments (chlorophyll a, chlorophyll b, carotenoids), electron carriers, and ATP synthase enzymes. The stroma is the fluid-filled matrix surrounding the grana, containing enzymes for the Calvin cycle, DNA, ribosomes, starch grains, and lipid droplets.



Light-Dependent Reactions

The light-dependent reactions occur on the thylakoid membranes and require light energy. The key events are:

1. **Photoexcitation:** Light energy is absorbed by chlorophyll molecules in photosystem II (PSII). Electrons in chlorophyll are excited to a higher energy level and are captured by electron acceptors.
2. **Photolysis of water:** The lost electrons are replaced by splitting water molecules (photolysis): $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. This is the source of the oxygen released in photosynthesis.
3. **Electron transport chain:** Excited electrons pass along a chain of electron carriers in the thylakoid membrane, losing energy at each step. This energy is used to pump H^+ ions from the stroma into the thylakoid space, creating a proton gradient.
4. **Chemiosmosis:** H^+ ions flow back to the stroma through ATP synthase (down the electrochemical gradient), driving the synthesis of ATP from ADP + Pi. This is called photophosphorylation.
5. **Photosystem I (PSI):** Electrons from the ETC are re-energised by light in PSI and transferred to $NADP^+$ reductase, which combines them with H^+ ions to reduce $NADP^+$ to NADPH (reduced NADP).

The products of the light-dependent reactions are ATP, NADPH (both used in the Calvin cycle), and O_2 (released as a waste product).

Light-Independent Reactions (Calvin Cycle)

The Calvin cycle occurs in the stroma and does not directly require light, but it depends on the ATP and NADPH produced by the light-dependent reactions. The key steps are:

1. **Carbon fixation:** CO_2 combines with the 5-carbon acceptor molecule ribulose bisphosphate (RuBP), catalysed by the enzyme RuBisCO. This produces an unstable 6-carbon intermediate that immediately splits into two molecules of 3-carbon glycerate-3-phosphate (GP).
2. **Reduction:** GP is reduced to glyceraldehyde-3-phosphate (G3P, also called GALP or triose phosphate) using ATP and NADPH from the light reactions. ATP provides energy; NADPH provides reducing power (hydrogen).
3. **Regeneration of RuBP:** Most G3P molecules (5 out of every 6) are used to regenerate RuBP using ATP. This keeps the cycle running. One in six G3P molecules is the net product, used to synthesise glucose, amino acids, lipids, or other organic molecules.

Exam Tip: For every 3 CO_2 molecules that enter the Calvin cycle, 6 G3P are produced but 5 are recycled to regenerate 3 RuBP. Only 1 G3P is the net gain. It takes 6 turns of the cycle (6 CO_2) to produce one glucose molecule (6 carbons = 2 × G3P combined).

5.2 Ecosystems and Energy Transfer

Ecosystem Concepts

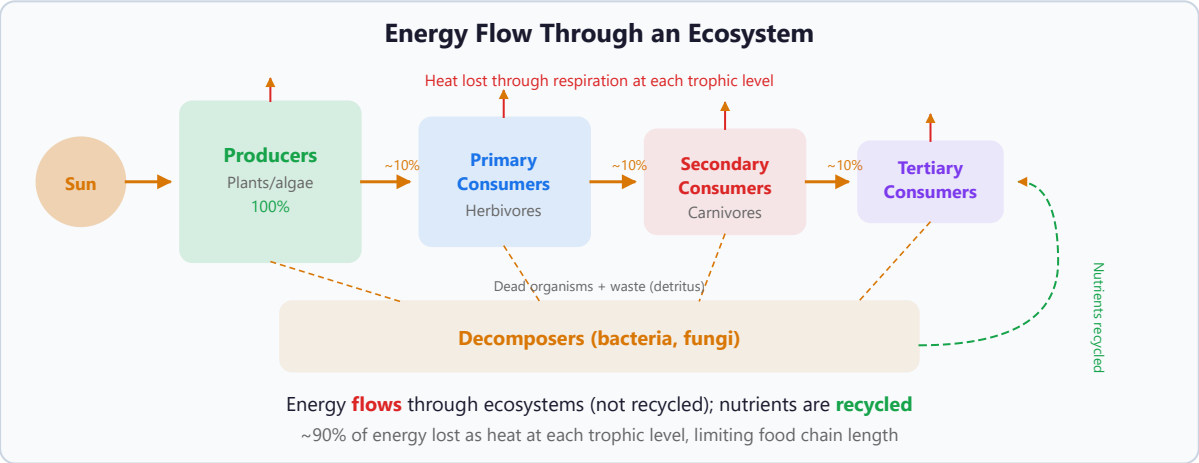
Ecosystem: A community of organisms (biotic component) interacting with each other and with their physical environment (abiotic component) in a defined area. Ecosystems are characterised by energy flow and nutrient cycling.

Key terms: a **habitat** is the place where an organism lives; a **niche** is the role of an organism within its ecosystem (what it eats, when it is active, where it lives, how it interacts with other species); a **population** is all the individuals of one species in a habitat; a **community** is all the populations of different species in a habitat.

Energy Flow Through Ecosystems

Energy enters most ecosystems as sunlight and is fixed by producers (autotrophs) through photosynthesis. Energy flows through the ecosystem via food chains and food webs: producers → primary consumers → secondary consumers → tertiary consumers. At each trophic level, energy is lost through:

- **Respiration:** The largest loss. Energy is released as heat during metabolic processes and cannot be recycled.
- **Excretion and egestion:** Not all ingested material is assimilated. Faeces contain undigested material; urine contains nitrogenous waste.
- **Death:** Dead organisms and waste are broken down by decomposers (bacteria and fungi), releasing energy as heat.



Typically only about 10% of energy is transferred from one trophic level to the next (though this varies: endotherms lose more energy through respiration for thermoregulation than ectotherms, so transfer efficiency is often lower in mammalian food chains). This limits food chains to usually 4-5 trophic levels and explains why there is less biomass at higher trophic levels (ecological pyramids of energy are always upright). It also explains why there are fewer top predators than herbivores in any ecosystem, and why food chains in aquatic ecosystems can sometimes be longer (ectothermic organisms lose less energy as heat).

Real-world application: The low efficiency of energy transfer between trophic levels has significant implications for food security. Producing 1 kg of beef requires approximately 8-10 kg of grain feed, because most of the energy in the grain is lost through the cow's respiration, movement, and metabolic processes. This is why eating lower on the food chain (more plant-based diets) is more energy-efficient and can support larger human populations. Intensive farming practices aim to increase energy transfer

efficiency by restricting animal movement (reducing respiratory losses) and controlling temperature (reducing thermoregulation costs), though these raise animal welfare concerns.

$$\text{Efficiency of energy transfer} = (\text{energy available at next trophic level} / \text{energy available at previous trophic level}) \times 100\%$$

Gross and Net Primary Productivity

$$\text{NPP} = \text{GPP} - \text{R}$$

Gross primary productivity (GPP) is the total rate at which energy is incorporated into organic molecules by producers through photosynthesis. Net primary productivity (NPP) is the rate at which energy is stored in plant biomass after respiratory losses are subtracted. NPP represents the energy available to primary consumers.

5.3 Climate Change

The Greenhouse Effect

The greenhouse effect is a natural phenomenon that maintains Earth's temperature at a habitable level. Short-wave radiation from the Sun passes through the atmosphere and warms the Earth's surface. The surface re-emits longer-wave infrared radiation, some of which is absorbed by greenhouse gases (CO_2, methane, water vapour, nitrous oxide) and re-radiated back to Earth, warming the atmosphere.

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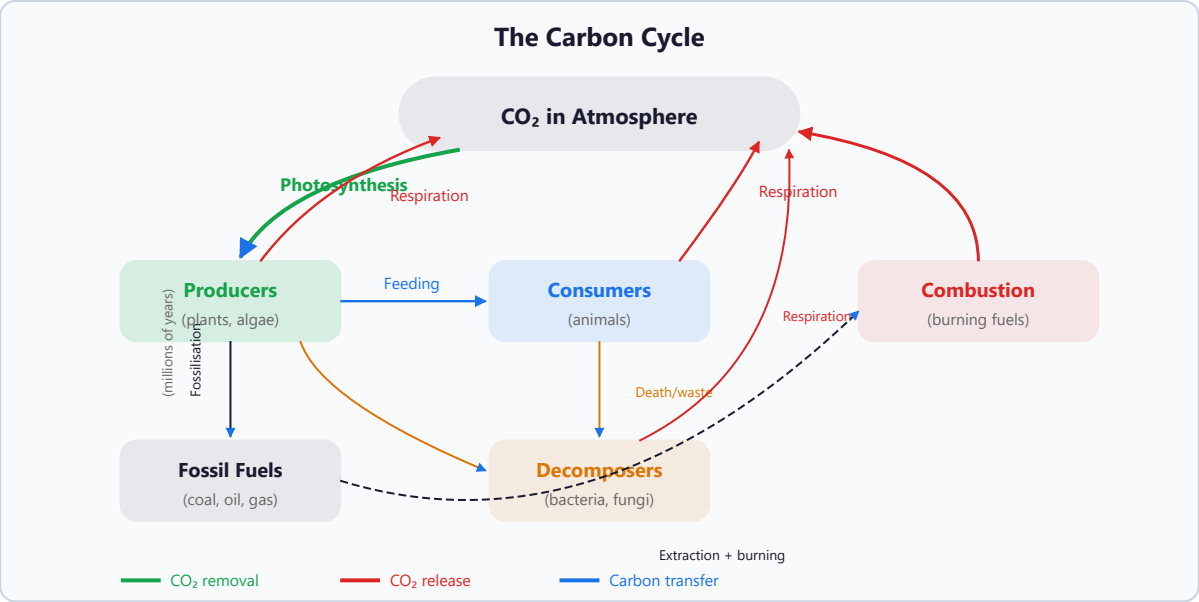


The **enhanced greenhouse effect** refers to the additional warming caused by increased concentrations of greenhouse gases due to human activities, primarily:

- **Burning of fossil fuels:** Releases CO_2 that was locked away in geological carbon sinks for millions of years, adding net carbon to the active carbon cycle. Atmospheric CO_2 has risen from ~280 ppm pre-industrial to over 420 ppm.
- **Deforestation:** Reduces CO_2 absorption by photosynthesis and releases stored carbon when trees are burned or decompose. Tropical deforestation accounts for ~10% of global emissions.
- **Agriculture:** Methane (CH_4) from enteric fermentation in ruminant livestock and anaerobic conditions in rice paddies. Methane is ~80 times more potent as a greenhouse gas than CO_2 over 20 years, though it persists in the atmosphere for a shorter time (~12 years vs centuries for CO_2).
- **Nitrous oxide (N_2O):** From agricultural fertilisers and industrial processes. ~300 times more potent than CO_2 per molecule.

Positive feedback loops may accelerate climate change: warming melts permafrost, releasing trapped methane; reduced ice cover decreases albedo (reflectivity), so more solar radiation is absorbed; warmer oceans absorb less CO_2 (gases are less soluble in warm water). These feedback mechanisms mean that initial warming can trigger further warming, potentially leading to tipping points beyond which change becomes self-reinforcing and irreversible.

Evidence for Climate Change

- **Ice core data:** Air bubbles trapped in ice cores from Antarctica and Greenland provide a record of atmospheric CO_2 and temperature going back hundreds of thousands of years. There is a strong correlation between CO_2 concentration and temperature.
- **Temperature records:** Global average temperatures have risen by approximately 1.1°C since pre-industrial times, with the rate of warming accelerating since the mid-20th century.
- **Sea level rise:** Thermal expansion of seawater and melting of ice caps and glaciers.
- **Retreating glaciers and sea ice:** Satellite measurements show significant losses.
- **Phenological changes:** Earlier flowering, earlier migration, shifts in species ranges toward the poles or higher altitudes.
- **Dendrochronology:** Tree ring analysis showing changes in growth patterns.
- **Pollen in peat bogs:** Changes in plant communities over time.

Consequences of Climate Change

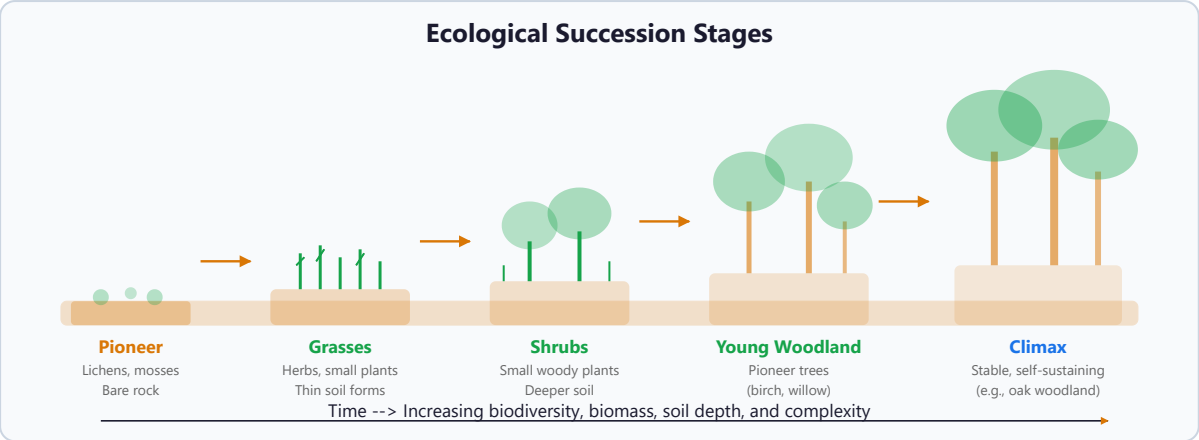
Climate change affects biodiversity through habitat loss (coral bleaching, sea level rise), disruption of ecological relationships (mismatched timing between species, e.g., caterpillars emerging before birds breed), range shifts, increased extreme weather events, and ocean acidification (CO_2 dissolving to form carbonic acid, affecting shell-forming organisms).

Exam Tip: Distinguish between the natural greenhouse effect (essential for life) and the enhanced greenhouse effect (caused by human activity). When discussing evidence, always evaluate its strengths and limitations (e.g., correlation does not prove causation, but the mechanism is well understood).

5.4 Ecological Succession

Succession

Succession: The gradual and directional change in the species composition of a community over time, eventually leading to a stable climax community. Primary succession begins on bare, previously uncolonised substrate; secondary succession occurs after disturbance of an existing community.



Pioneer species (e.g., lichens, mosses) are the first to colonise bare rock. They are adapted to harsh conditions and help create soil by weathering rock and adding organic matter when they die. As soil develops, more complex plant communities can establish, each seral stage modifying the environment to make it more suitable for the next stage. Eventually, a climax community is reached (determined by climate and soil type), which is relatively stable and self-sustaining.

5.5 Evolution and Speciation

Natural Selection

> **Natural selection:** The mechanism of evolution by which organisms with advantageous traits are more likely to survive, reproduce, and pass their alleles to the next generation. Over time, the frequency of beneficial alleles increases in the population.

The conditions for natural selection are: genetic variation within a population (caused by mutation, meiosis, and sexual reproduction); competition for limited resources (overproduction of offspring); differential survival and reproduction (organisms with favourable traits are more likely to survive); and inheritance (advantageous traits are passed to offspring).

Types of Selection

- **Directional selection:** Individuals at one extreme of a phenotypic range have greater fitness. The population mean shifts in one direction. Example: peppered moths during industrial pollution (darker moths had higher survival).

- **Stabilising selection:** Individuals with intermediate phenotypes have greatest fitness; extremes are selected against. Reduces variation but does not change the mean. Example: human birth weight (very small or very large babies have lower survival).
- **Disruptive selection:** Individuals at both extremes have greater fitness than those in the middle. Can lead to speciation. Example: beak size in seed-eating birds on islands with only very large or very small seeds.

Speciation

Speciation: The formation of new species from an existing population, occurring when populations become reproductively isolated so that gene flow ceases, and they diverge genetically until they can no longer interbreed to produce fertile offspring.

Allopatric speciation: Geographical isolation separates a population (e.g., mountain range, river, island colonisation). The isolated populations experience different selection pressures and genetic drift, accumulating genetic differences until they become reproductively incompatible.

Sympatric speciation: New species arise within the same geographical area without physical barriers. Mechanisms include polyploidy (especially in plants, where chromosome doubling creates instant reproductive isolation), ecological specialisation, or temporal isolation (breeding at different times).

Evidence for Evolution

Evidence includes the fossil record (showing change over time and transitional forms), comparative anatomy (homologous structures indicating common ancestry), molecular biology (DNA and protein sequence similarities), biogeography (distribution of species reflecting evolutionary history), and direct observation of evolution (antibiotic resistance in bacteria, pesticide resistance in insects).

Exam Tip: When describing natural selection, use precise language: variation exists, those with advantageous traits are more likely to survive and reproduce, these traits are inherited, and allele frequency changes over generations. Avoid Lamarckian language (organisms do not "choose" to evolve or "develop" traits because they need them).

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